Sustainable Development Analytics

YEAR- II (SEMESTER- III)

ASSIGNMENT

SDG Dashboard:

SUBMITTED TO-

Dr. Rahul S.

SUBMITTED BY-

Aditi Ranjan (M2024BSASS003)

Table of Contents

Six Pillar Transformations Accelerator

Dashboard Link: https://m2024bsass003.shinyapps.io/sdgaccelarator/

The tool is exploratory and policy-oriented, not predictive

About the dashboard:

This theme is inspired from the United Nations Sustainable Development Goals Report 2025. In that official UN report, the world leaders looked at progress on the 17 SDGs after 10 years (since 2015) and said: "We are not moving fast enough. Only about 35% of the targets are on track. Most goals are at risk if we continue like this."

To fix this, the report strongly recommended focusing on six key priority areas (also called "six transformations" or "six priority transformations"). These six areas were chosen because they can create big positive chain reactions (cascading benefits) across many other SDGs if we invest in them now. The six are:

Food systems: Making sure everyone has enough healthy food without harming the planet.

Energy access: Giving everyone clean, affordable energy (like electricity and renewables).

Digital transformation: Connecting people with internet and mobile technology.

Education : Improving quality learning for everyone.

Jobs and social protection: Creating decent work and safety nets for people who need help.

Climate and biodiversity: Fighting climate change and protecting nature (forests, oceans, animals).

The UN said these six are like "unlockers": when you improve one, it helps others too.

Data Source and Indicator Selection

The data was taken from the UN SDG Global database. Tier I indicators were selected for the analysis.

Time period covered: 2015-2023.

2015, reflecting the post-2015 SDG monitoring period.

Indicators: were selected accordingly for the Six Transformation Frameworks.

For each one of the transformation pillars, one or more indicators were taken to capture core outcome dimensions.

Food systems: Prevalence of undernourishment (%)

Energy access: Proportion with access to electricity (%), Proportion with primary reliance on clean fuels/technology (%), Renewable energy share in total final consumption (%)

Digital transformation: Proportion of population covered by at least a 4G mobile network (%)

Education: Completion rate (lower secondary) (%)

Jobs and social protection: Unemployment rate, by sex and age (%), Proportion of youth not in education/employment/training (%), Proportion of population covered by at least one social protection benefit (%)

Climate and Biodiversity: Total greenhouse gas emissions excluding LULUCF (Mt CO2e), Forest area as a proportion of total land area (%)
For GHG emissions very sparse data was given so data was added from world bank dataset.

Data Cleaning and Preparation

Before transformation, the dataset was cleaned to ensure consistency across indicators and years. Missing values were not imputed using statistical interpolation or replacement. Instead, missing observations were retained as NA and handled explicitly during aggregation steps using na.rm = TRUE. This approach avoids introducing artificial trends or assumptions into the data and ensures transparency about data gaps.
Importantly, a country was not dropped entirely if some indicators were missing for certain years. Instead, the latest available year for each indicator was used independently when constructing the radar chart, which is explicitly communicated in the dashboard annotation (e.g., "Education: 2020, Climate: 2023").

Analytical Framework

Normalisation

To ensure comparability across indicators with different units and ranges, all indicators were normalised to a 0–100 scale using a global, percentile-based approach.
For each indicator i, lower (2.5th percentile was used) and upper bounds(97.5th percentile) were used and calculated across all countries and years. These bounds were fixed and global thus making normalisation parameters same across countries.
Because some indicators improve when values increase (e.g., electricity access), while others improve when values decrease (e.g., greenhouse gas emissions), a direction-aware normalisation function was applied.

Positive-direction indicators:

$$Score_i = \frac{(value - lower)}{(upper - lower} * 100$$

Negative-direction indicators:

$$Score_i = \frac{(Upper - value)}{(upper - lower} * 100$$

All scores were capped within [0, 100]:
Missing values remained as NA and were not imputed.

Aggregation into Six Transformations Score

Normalized indicators were then aggregated into the six transformation scores. Aggregation was done using simple arithmetic means, with equal weighting across indicators within each transformation. For example, the Energy transformation was computed as the mean of normalized electricity access, clean cooking and renewable energy indicators, while Climate was calculated as the mean of normalized greenhouse gas emissions and forest area.

Overall Momentum Score Construction

An Overall Momentum Score was computed for each country-year as the mean of the six transformation scores. This score represents a synthetic measure of system-wide SDG progress, rather than performance in any single sector. As with transformation-level aggregation, equal weighting was used to avoid privileging one transformation over others.

Correlation-Based Spillover Matrix Construction

For each pair of transformations, Pearson correlation coefficients were calculated using historical country-year data. These correlations capture empirical co-movement patterns between transformations over time and across countries. The resulting correlation matrix forms a spillover weight matrix, where each element represents the strength of association between two transformations.

These correlations are non-causal. They do not imply that improvements in one transformation mechanically cause improvements in another. Instead, they serve as empirical proxies for systemic interlinkages.
Negative or weak correlations naturally result in smaller or negligible spillover effects, while stronger positive correlations lead to more pronounced cross-sector transmission.

Policy Acceleration Level %

It captures the intensity of coordinated policy effort, institutional reform, implementation capacity, and governance effectiveness applied to a selected transformation.
The boost operates by increasing the remaining gap to the performance frontier (defined as a score of 100), rather than adding a fixed amount. This ensures diminishing returns and avoids unrealistic jumps for high-performing countries.
For a selected transformation, the boosted score is calculated as:

Boosted Score = Current Score + ß * (100 - Current Score)
where:

- ß is the boost factor derived from the user-selected boost percentage
- (100 - Current Score) represents the remaining distance to the frontier.

All boosted scores are capped at 100 to preserve interpretability.

Spillover-Adjusted Boost Transmission

The same boost logic is extended to non-targeted transformations through correlation-based spillovers. For any other transformation j, the spillover gain is computed as:

Spillover Gain = $Correlation_{f\,j} *$ ß $* (100 - Current\ score_j$

where: \text{Correlation}_{f,j} is the correlation coefficient between the selected (focused) transformation f and transformation j,

ß = Policy Acceleration Boost (%) /100

(100 - $Current\ score_j$) is the remaining distance to the maximum achievable score.

The final boosted score for transformation j is then:

Boosted Score = $Current\ score_j$ + $Spillover\ gain_j$

All boosted scores are capped at 100.

Overdrive Mode (High-Coordination Policy Scenario)

Overdrive Mode represents a scenario of system-wide policy coordination, where governance, finance, institutions and implementation capacity act together across sectors. It amplifies the strength of policy transmission in the simulation. This mode is exploratory and scenario-based and not predictive.

Overdrive is implemented by scaling up the boost factor:
Overdrive Boost Factor = 1.5 * Boost Factor

This amplified boost factor is then applied in both the direct boost and spillover equations:
Boosted Score = Current Score + Overdrive Boost Factor * (100 - Current Score)

and for spillovers:

$Spillover\ gain_j$= $Correlation_{f\ j}$ * Overdrive Boost Factor * (100 - $Current\ score_j$

All resulting scores are capped at 100.

Key insights

This dashboard provides exploratory insights into how progress across SDG-related sectors is interconnected rather than independent. The analysis suggests that accelerating action in one transformation often coincides with improvements in others, highlighting the presence of strong cross-sector linkages within the SDG framework
Food and Energy emerge as frequent high-impact entry points, showing relatively larger spillover gains for Jobs, Education and Climate outcomes in several countries.
The results also suggest that coordinated policy action can amplify system-wide progress.
Importantly, these findings are scenario-based and exploratory. They do not predict future outcomes but instead illustrate how different policy intensities and coordination assumptions could shape SDG trajectories under plausible conditions

Policy relevance of findings

This dashboard is policy-relevant because it shows that progress toward the SDGs is interconnected. Improving one sector often leads to gains in others, meaning policies should not be designed in isolation.

The findings also show that policy coordination matters. When actions across sectors are aligned (as represented by Overdrive Mode), overall SDG progress increases more than with separate, incremental efforts. This suggests that integrated planning, cross-departmental coordination and aligned investments are crucial for achieving the SDGs by 2030.

Assumptions and Limitations

Assumptions

Several assumptions were required to operationalise the SDG data into an interactive policy simulation dashboard.

First, it is assumed that the selected Tier I SDG indicators adequately represent each transformation domain. While each transformation is complex and multidimensional, only a limited number of indicators were used due to data availability and comparability constraints. The analysis assumes that changes in these indicators reasonably reflects broader progress within each transformation.

Second, all indicators were normalised to a common 0-100 scale using percentile-based normalisation. This assumes that relative performance across countries is more informative for policy comparison than absolute indicator values. As a result, the dashboard reflects relative position and progress, rather than the absolute achievement of SDG targets.

Third, when constructing transformation scores, the latest available year for each indicator was used, even if different indicators correspond to different years. This assumes that the most recent observation best captures current momentum, and that small timing differences do not substantially distort the overall transformation score.

Fourth, the aggregation of indicators within each transformation assumes equal importance. No explicit weighting scheme was applied because there is no universally accepted method for assigning weights to SDG indicators. Equal weighting was therefore chosen as a transparent and neutral assumption.

Fifth, the spillover effects between transformations are assumed to be symmetric and proportional to historical correlations. Correlation coefficients were used as proxies for cross-sector linkages, under the assumption that stronger observed correlations reflect stronger potential policy spillovers. This approach does not imply causality but provides a structured way to model interdependencies.

Sixth, the Investment Boost parameter is assumed to represent policy effort intensity rather than monetary investment. The boost is interpreted as an increase in coordinated policy focus, regulatory effort, and institutional capacity, not financial expenditure.

Finally, Overdrive Mode assumes a high-coordination policy environment, where governance, institutions and implementation capacity act simultaneously across sectors. The model assumes that such coordination amplifies spillover effects uniformly across transformations.

Limitations

First, the analysis is scenario-based and exploratory, not predictive. The boosted and overdrive scenarios illustrate what could happen under stronger policy coordination, but they do not forecast actual future outcomes.

Second, the use of correlation-based spillovers does not establish causality. A positive spillover from one transformation to another should not be interpreted as a guaranteed or direct causal relationship. Structural, institutional and country-specific factors may weaken or reverse such relationships in practice.

Third, data availability varies significantly across countries and indicators. Some countries have sparse reporting for certain transformations, which may affect the robustness of their scores and spillover responses. Although countries with severe data gaps were excluded, residual inconsistencies remain.

Fourth, percentile normalisation can mask absolute performance differences. A country may rank relatively well despite still being far from achieving SDG targets in absolute terms. Conversely, countries with strong absolute performance may show smaller gains due to already high baseline values.

Fifth, the linear trend projection to 2030 assumes that past trends will continue. This does not account for structural breaks, economic shocks, pandemics, conflicts, or major policy reforms that could significantly alter trajectories.

Sixth, the Climate transformation combines indicators with different directional meanings, such as greenhouse gas emissions (where lower values are better) and forest area (where higher values are better). Although normalisation corrects for direction, the combined score may still oversimplify complex environmental dynamics.

Finally, the model does not account for country-specific policy constraints, such as fiscal capacity, political economy or institutional quality. As a result, identical boost scenarios may not be equally feasible across countries.

Tools and Packages used

- R
- RStudio
- R shiny
- shinydashboard
- dplyr
- plotly
- shinyalert
- jsonlite

AI DECLARATION

AI tools were used solely to assist in organising, structuring and refining my analytical steps and reasoning into clear, grammatically correct written form.
All data processing, analysis, design and interpretations are of my own.